

# Facilitating Change: Implementation Science 101 for Medical Educators

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# Disclosures

- Association of Program Directors in Surgery Mark Friedell Research Grant
- Feinberg School of Medicine Augusta Webster Fellowship
- American Board of Medical Specialties Scholars Program
- American Board of Surgery



**Many thanks to Kelli Scott PhD!!**

# Overview



Discuss why implementation science is an important resource for medical educators



Define Implementation Science



Review some basic tools for Implementation Science



Work through an example together (in groups)

# Overview



Discuss why implementation science is an important resource for medical educators

# Why should I care about implementation science as a medical educator?

- As medical educators, you have likely witnessed innovations or processes that have been implemented unsuccessfully
- **Unintended byproducts of unsuccessful implementation:** loss of trust, an increase in stakeholder burnout, and negative impacts on process improvement efforts
- Often, we instinctively move through an implementation science framework but don't slow down and give adequate attention to all the elements that influence successful adoption

# Overview

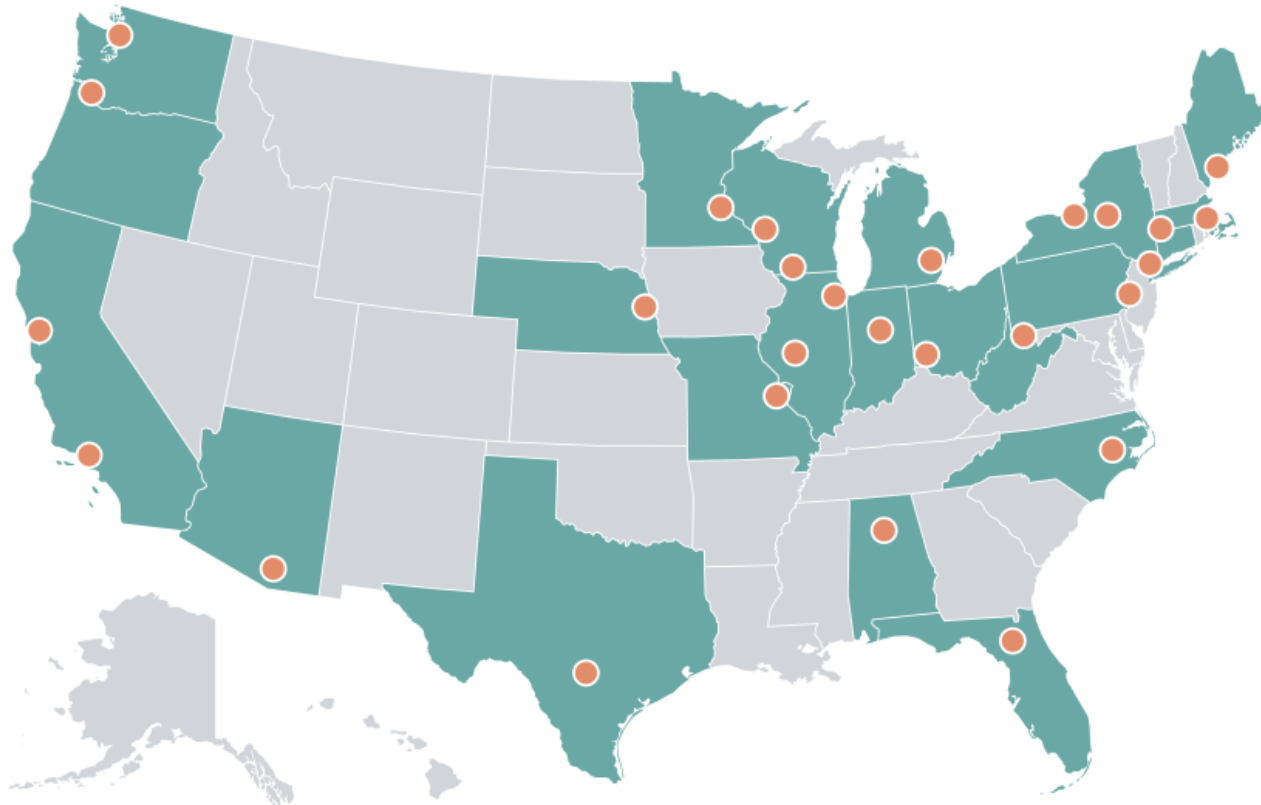


Discuss why implementation science is an important resource for medical educators

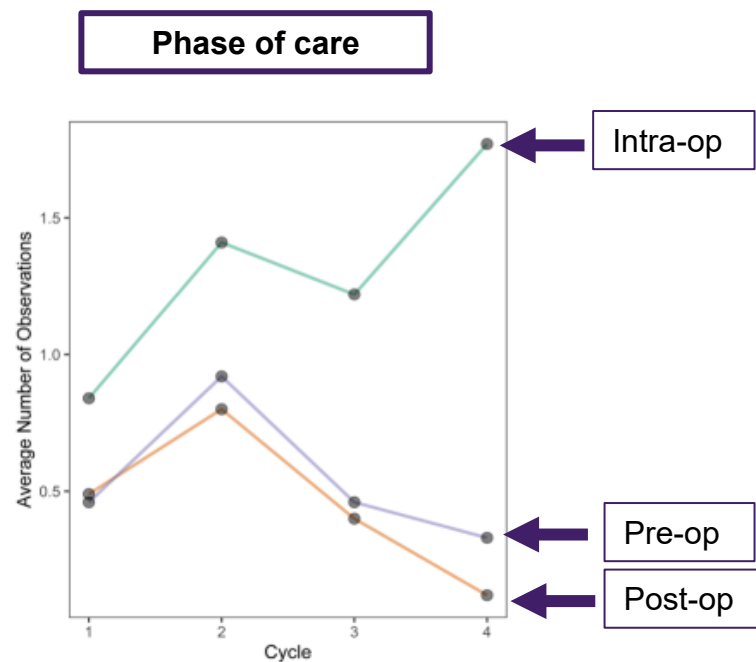
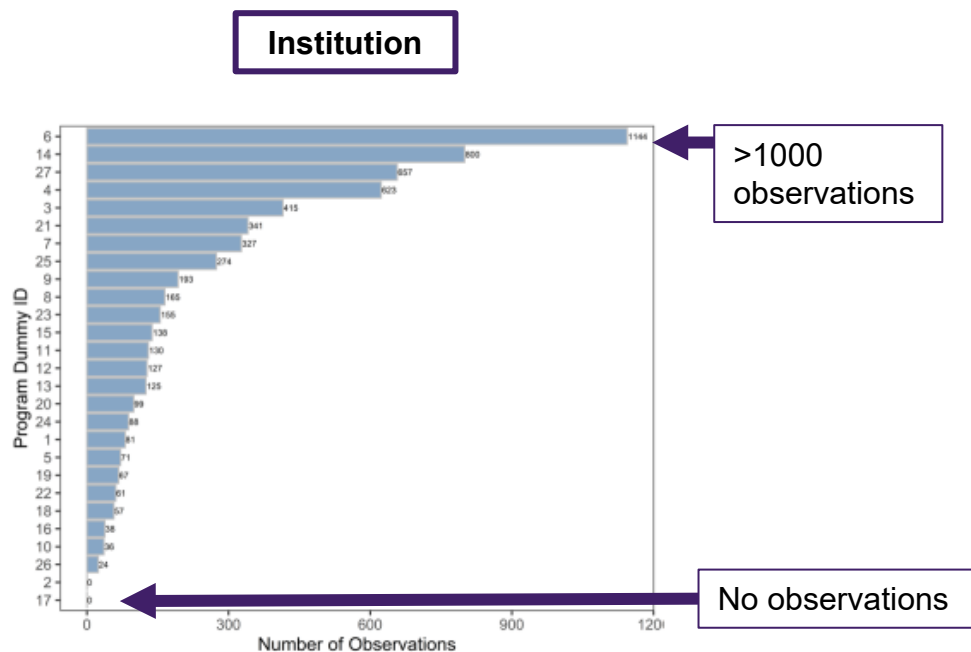


Define Implementation Science

## The ABS EPA pilot from 2018-2020 at 28 diverse residency programs



# Engagement with EPAs is highly variable



(Brasel et al, *Annals of Surgery*, 2023)

# Faculty and resident perspectives on EPA implementation

Faculty and Residents felt that EPAs were a **useful and valuable** addition to training

Faculty and residents thought EPAs were **NOT overly time consuming** to complete

Faculty and residents reported **difficulty incorporating** them into their daily workflow

(EPA Best Practices Committee: Williams-Karnesky et al, *J Surg Ed*, 2024)

People may **buy-in conceptually** to the idea of EPAs,  
and yet may not **enact the behavior** of completing  
EPAs

People may **buy-in conceptually** to the idea of EPAs,  
and yet may not **enact the behavior** of completing  
EPAs

*How do we get people to **DO THE  
THING?***

# Center for Dissemination & Implementation Science



Kelli Scott, PhD

# What is Implementation Science?

Implementation Science is the **scientific study** of **methods and strategies** that **facilitate the uptake** of evidence-based practice and research into **regular use**.

# Implementation science in (very) non-technical language

The intervention/practice/innovation is **THE THING**

*Implementation research* looks at how best to help people **DO THE THING**

*Implementation strategies* are the stuff we do to try to help people **DO THE THING**

Main *implementation outcomes* are **HOW MUCH** and **HOW WELL** people **DO THE THING**

(Curran, *Implementation Science Communications*, 2020)

# Overview



Discuss why implementation science is an important resource for surgical educators

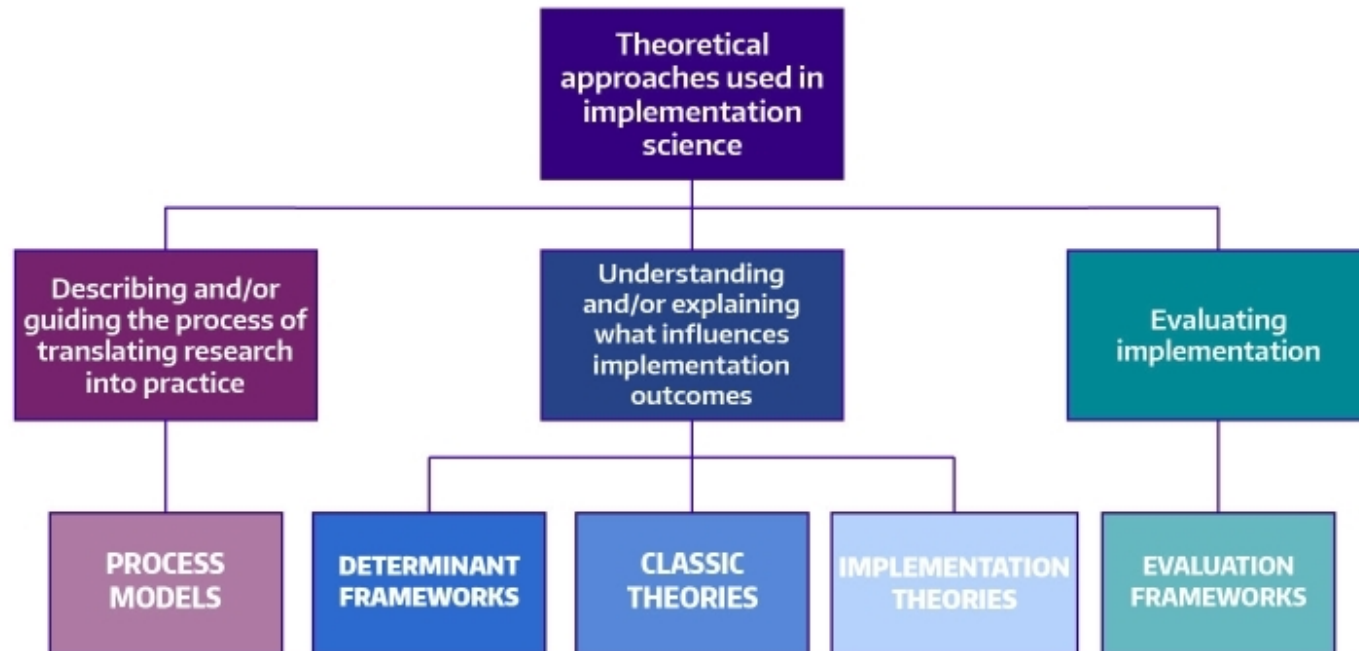


Define Implementation Science



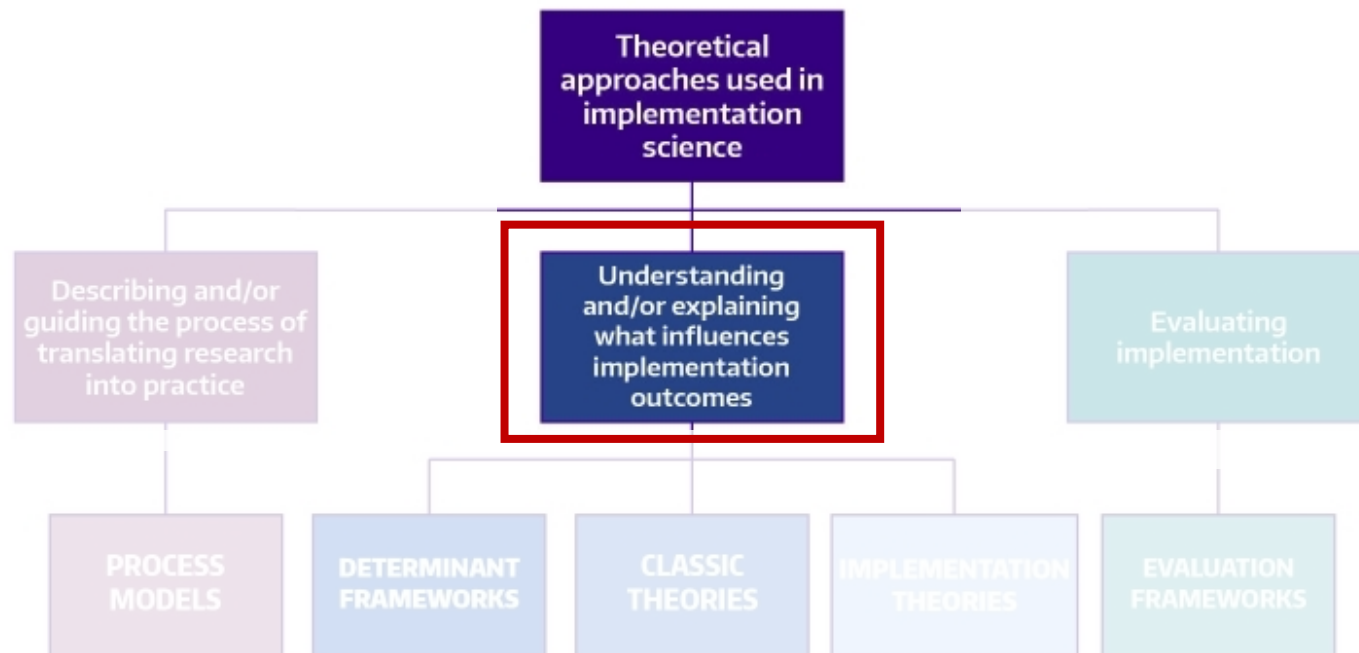
Review some basic tools for Implementation Science

# There are many Implementation Science strategies



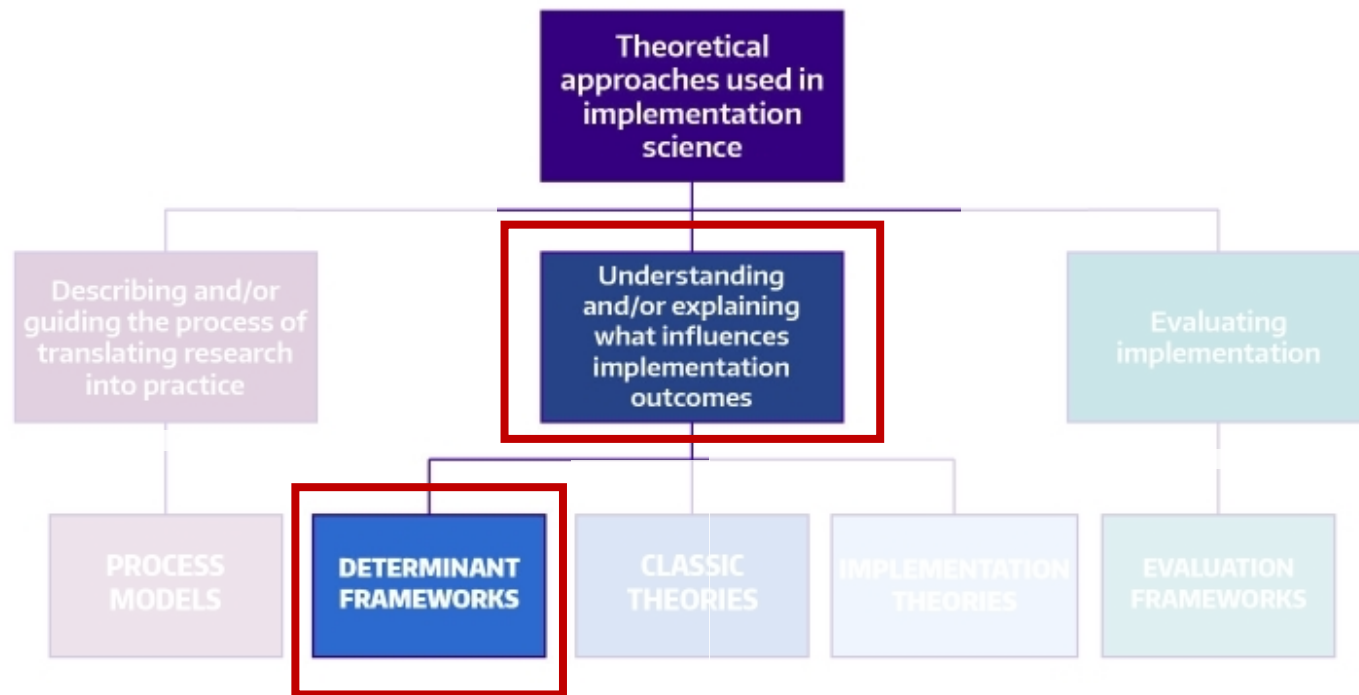
(Nilsen, *Implementation Science*, 2015)

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# There are many Implementation Science strategies



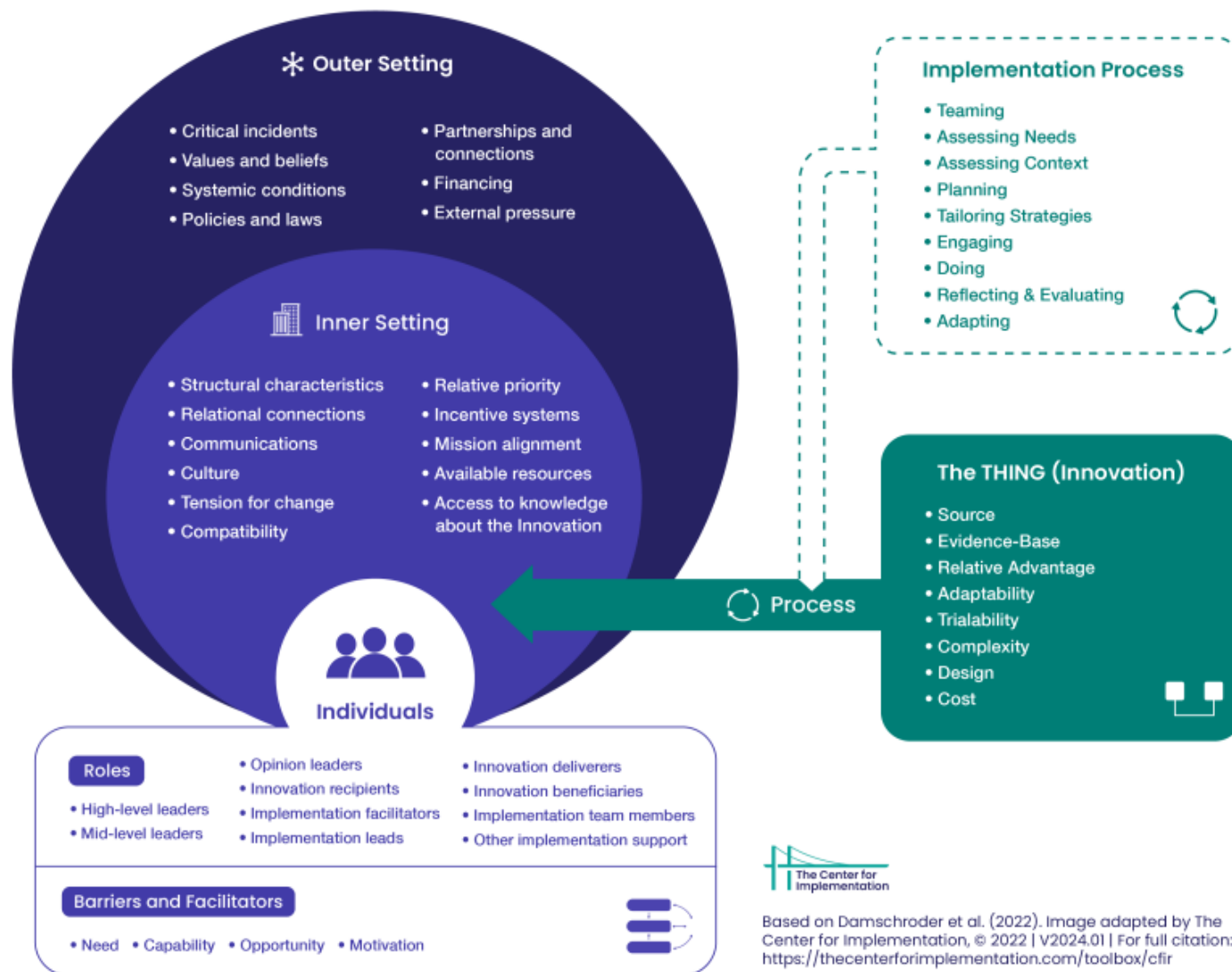
(Nilsen, *Implementation Science*, 2015)

# How can an Implementation Science framework help?

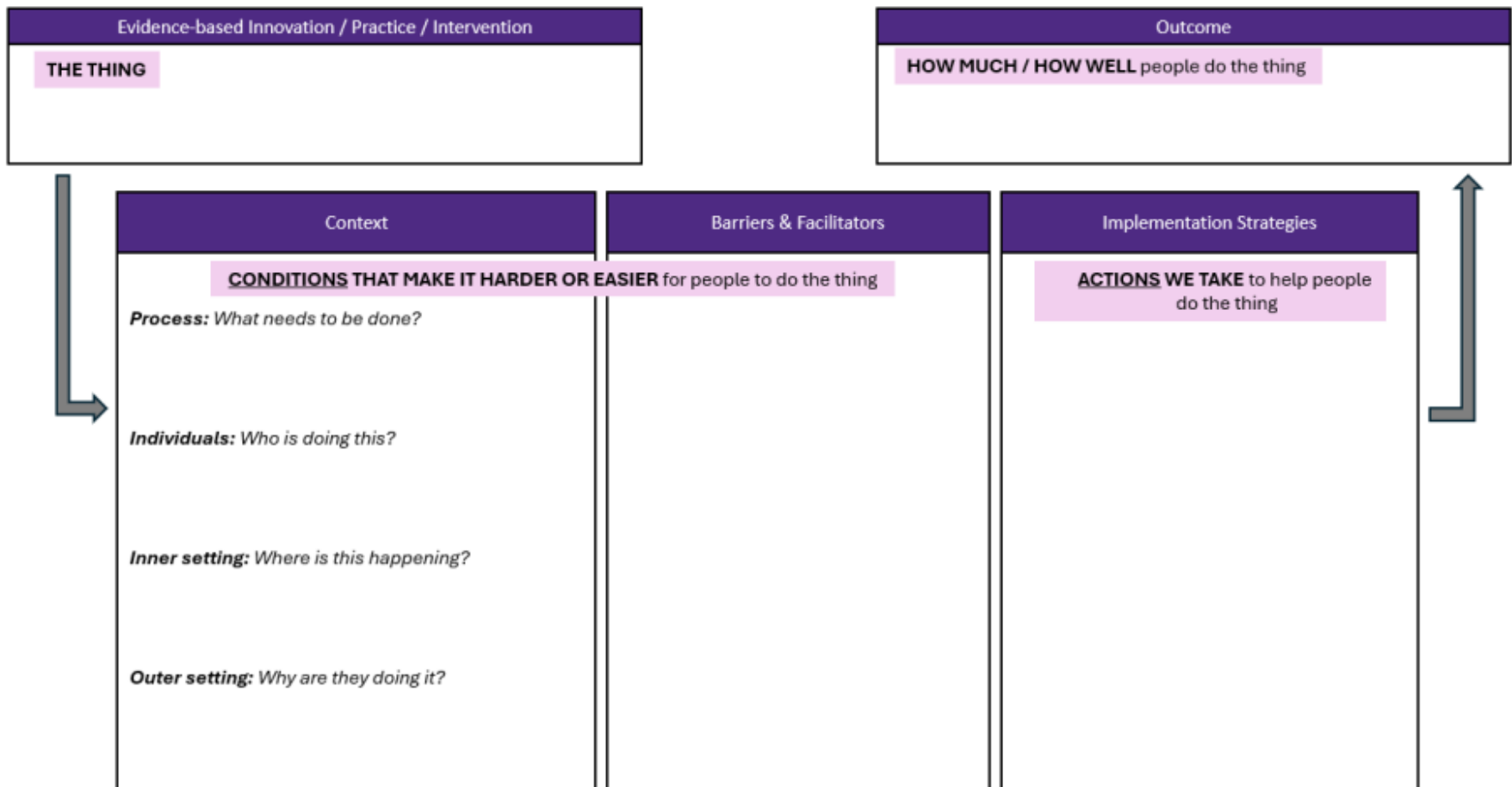
## Purpose:

To provide a **comprehensive** and **standardized approach** for understanding the complex factors that influence the **implementation and sustainability** of innovations.

# The Consolidated Framework for Implementation Research (CFIR) can be used to define the *context*



# An Implementation Logic Model (ILM) can be used to study the implementation *process*



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Work through an example together (in groups)

# Applying the tools of Implementation Science

Implementation Science can seem overwhelming and have a lot of “buzz words” **BUT** we can treat this like following *a new recipe*



The image shows a recipe card template. At the top, there is a header with the word "Recipe" in a cursive font, flanked by illustrations of kitchen items: a knife, a pan, a rolling pin, and a can. Below the header, there are several fields for recipe information:

- TITLE: \_\_\_\_\_
- PREP TIME: \_\_\_\_\_ | COOK TIME: \_\_\_\_\_ | SERVINGS: \_\_\_\_\_
- INGREDIENTS: \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_
- DIRECTIONS: \_\_\_\_\_  
\_\_\_\_\_  
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\_\_\_\_\_  
\_\_\_\_\_

# Your Implementation Science **RECIPE**

## **Step 1: Decide what to make**

- Name **THE THING**, define **outcomes**

## **Step 2: List the ingredients**

- Understand the **Context**, identify **Barriers and Facilitators**

## **Step 3: Mix & Bake**

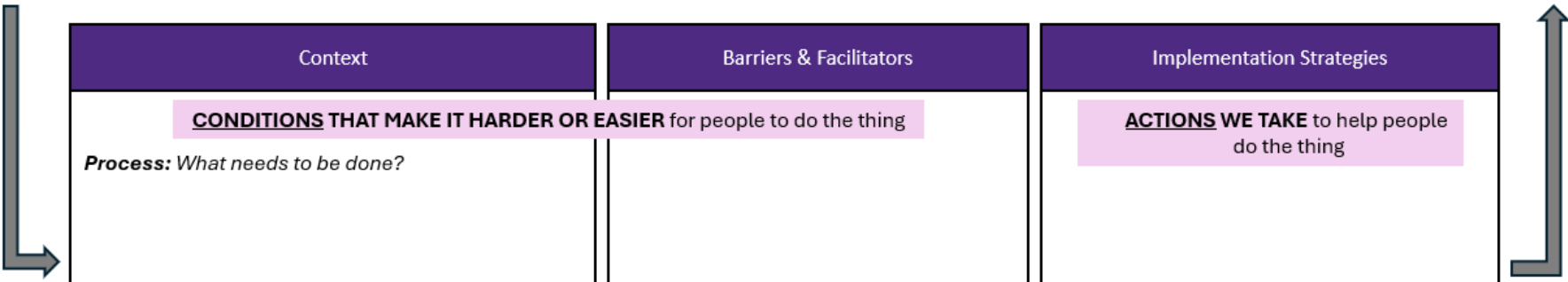
- Develop implementation **Strategies**

## **Step 4: Frost & Serve**

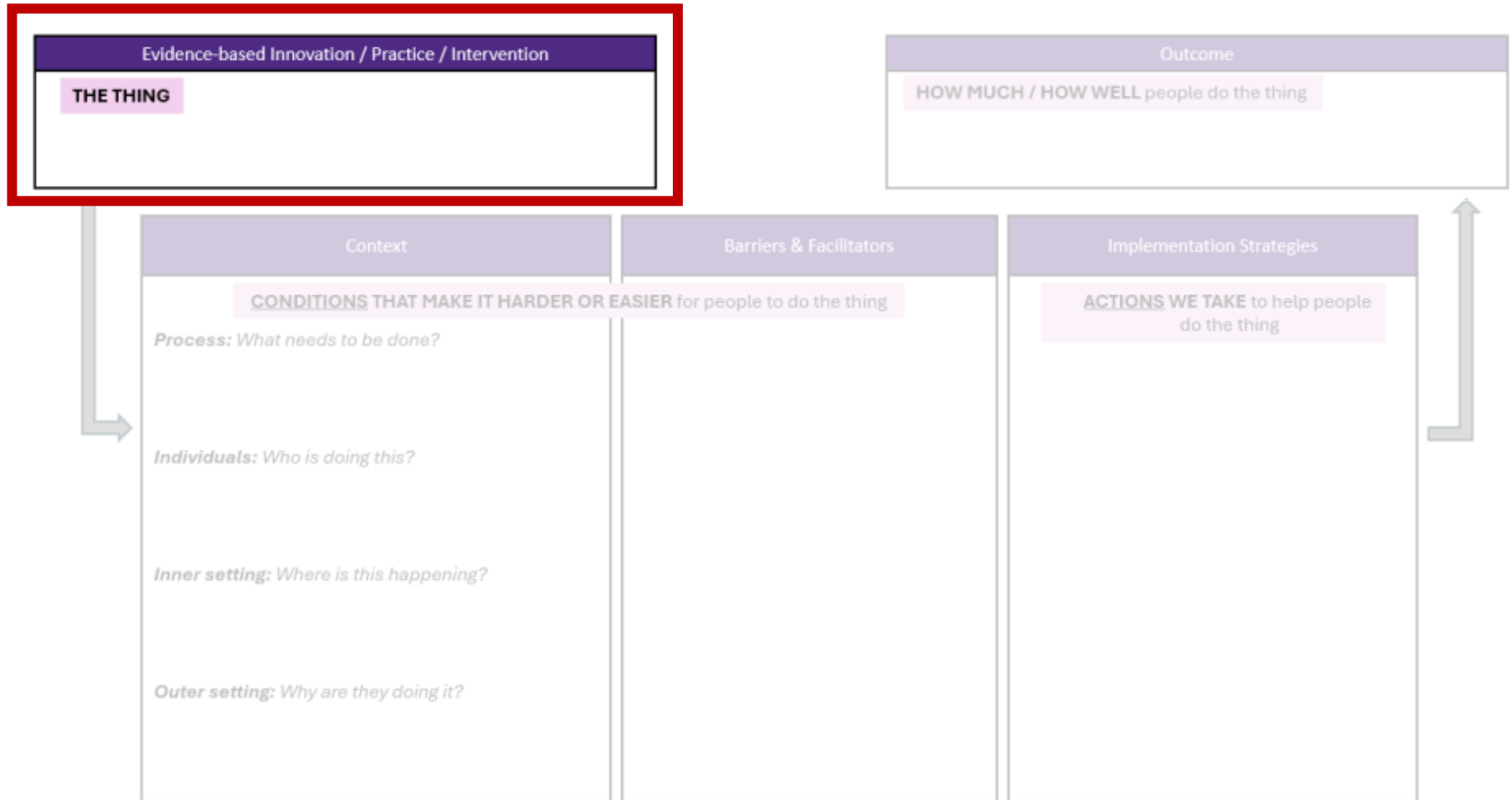
- Evaluate **Impact**

# Create your **Recipe** using the ILM

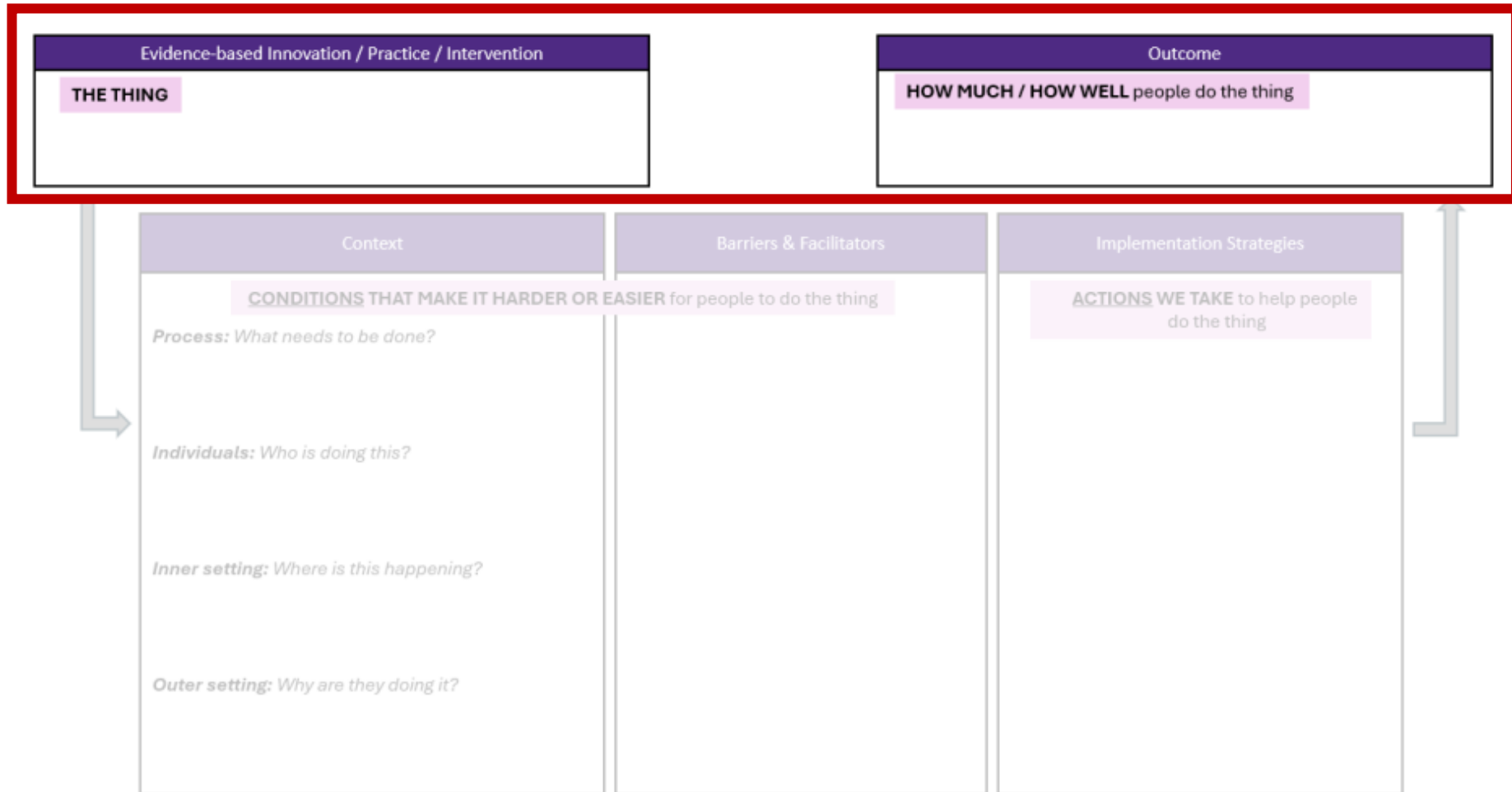
| Evidence-based Innovation / Practice / Intervention                               |                                    | Outcome                                                   |
|-----------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------|
| <b>THE THING</b>                                                                  |                                    | <b>HOW MUCH / HOW WELL</b> people do the thing            |
| <b>Context</b>                                                                    | <b>Barriers &amp; Facilitators</b> | <b>Implementation Strategies</b>                          |
| <b><u>CONDITIONS</u> THAT MAKE IT HARDER OR EASIER</b> for people to do the thing |                                    | <b><u>ACTIONS</u> WE TAKE</b> to help people do the thing |
| <i>Process: What needs to be done?</i>                                            |                                    |                                                           |
| <i>Individuals: Who is doing this?</i>                                            |                                    |                                                           |
| <i>Inner setting: Where is this happening?</i>                                    |                                    |                                                           |
| <i>Outer setting: Why are they doing it?</i>                                      |                                    |                                                           |



# Step 1: Decide what to make

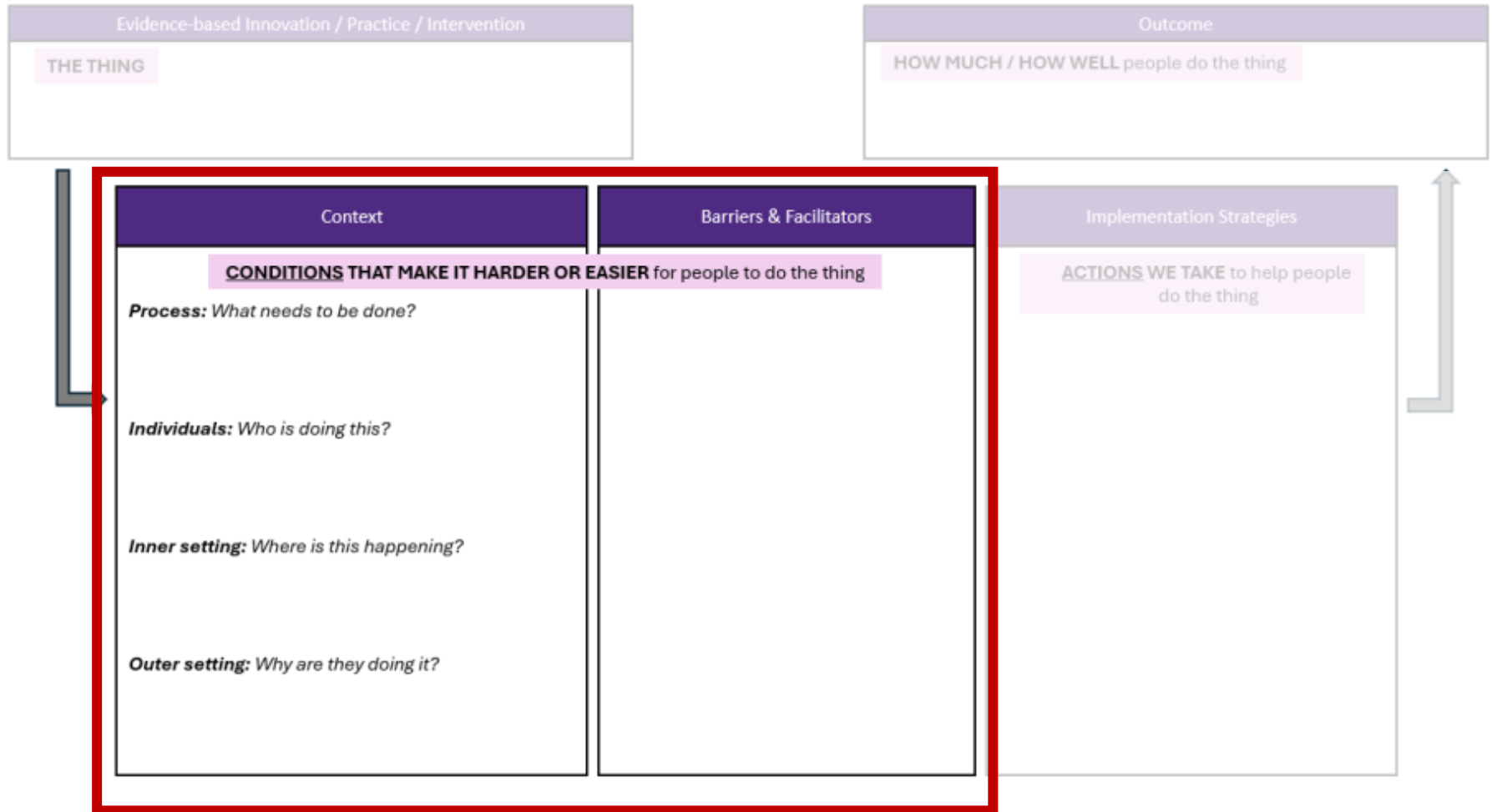


# Step 1: Decide what to make



# Step 2: List your ingredients

Consider the **Context** and identify **Barriers and Facilitators**



# Step 2: List your ingredients

Consider the **Context** and identify **Barriers and Facilitators**

- **Inner Setting:** Where the implementation process takes place (internal dynamics)
- **Outer Setting:** Broader environment in which the organization operates
- Important to pay attention to the synergy between the outer and inner setting



# Step 2: List your ingredients

Consider the **Context** and identify **Barriers and Facilitators**

## **Organizational Assessment**

- › What factors influence the readiness of an organization to implement a new intervention?
- › How do contextual factors within the organization affect the implementation process?
- › What is the role of organizational leadership in supporting or hindering implementation efforts?
- › How do inter-organizational relationships and networks influence the implementation process?

## **Stakeholder Analysis**

- › How do different stakeholders influence the implementation of evidence-based interventions?
- › What are the policy barriers and facilitators to implementing new interventions?
- › How can stakeholder engagement be optimized to enhance the implementation process?

## **Social Marketing**

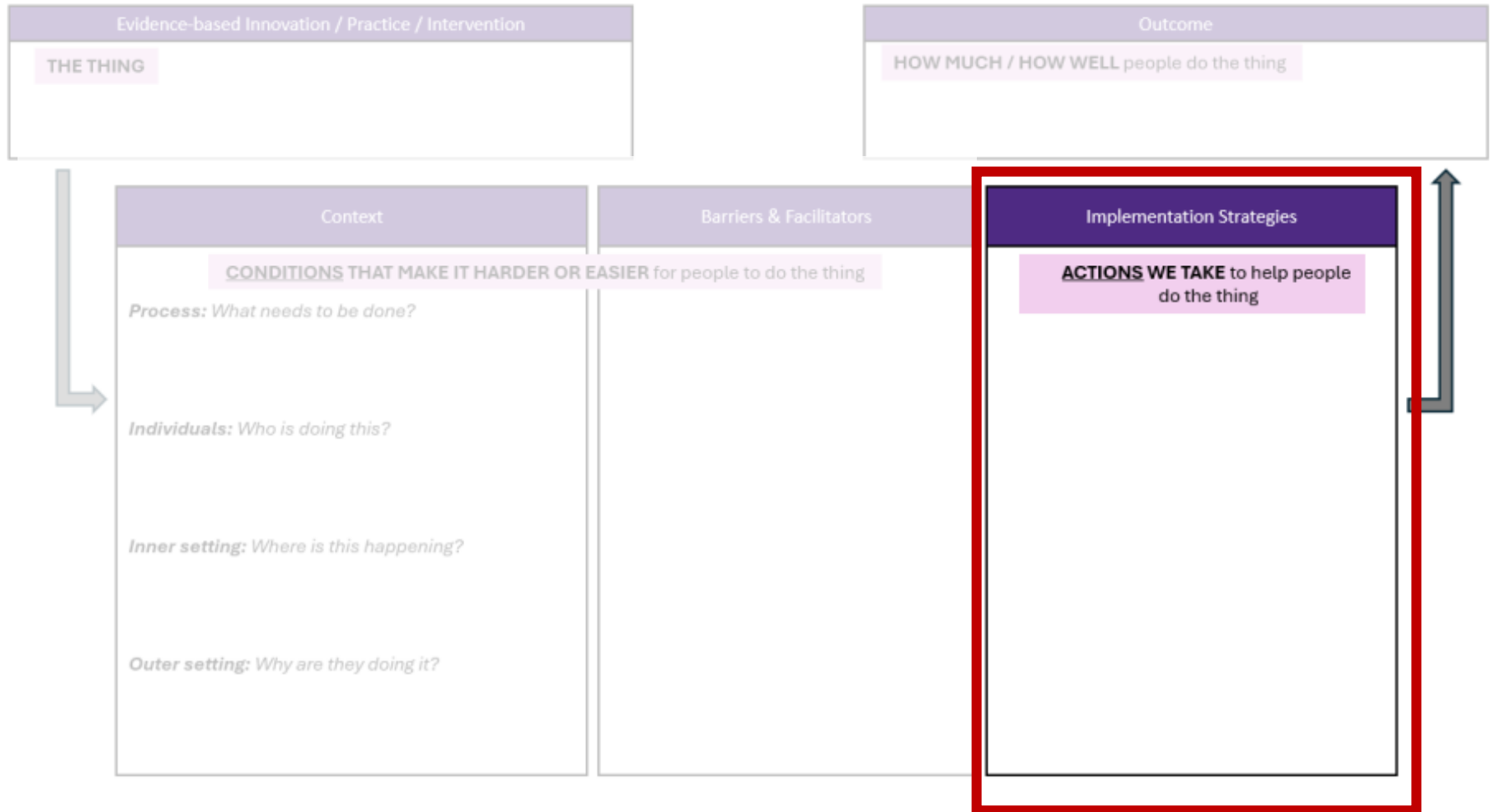
- › How can one effectively segment the target audience to tailor implementation strategies?
- › What are the most effective communication strategies to promote the adoption of evidence-based practices?
- › What are the barriers and facilitators to behavior change within the target population?

## **Prioritizing Implementation Barriers**

- › Which barriers have the greatest impact on the implementation process?
- › How do different barriers interact and compound to affect implementation outcomes?
- › What strategies are most effective in overcoming prioritized barriers?
- › How do contextual factors influence the prioritization and impact of barriers?

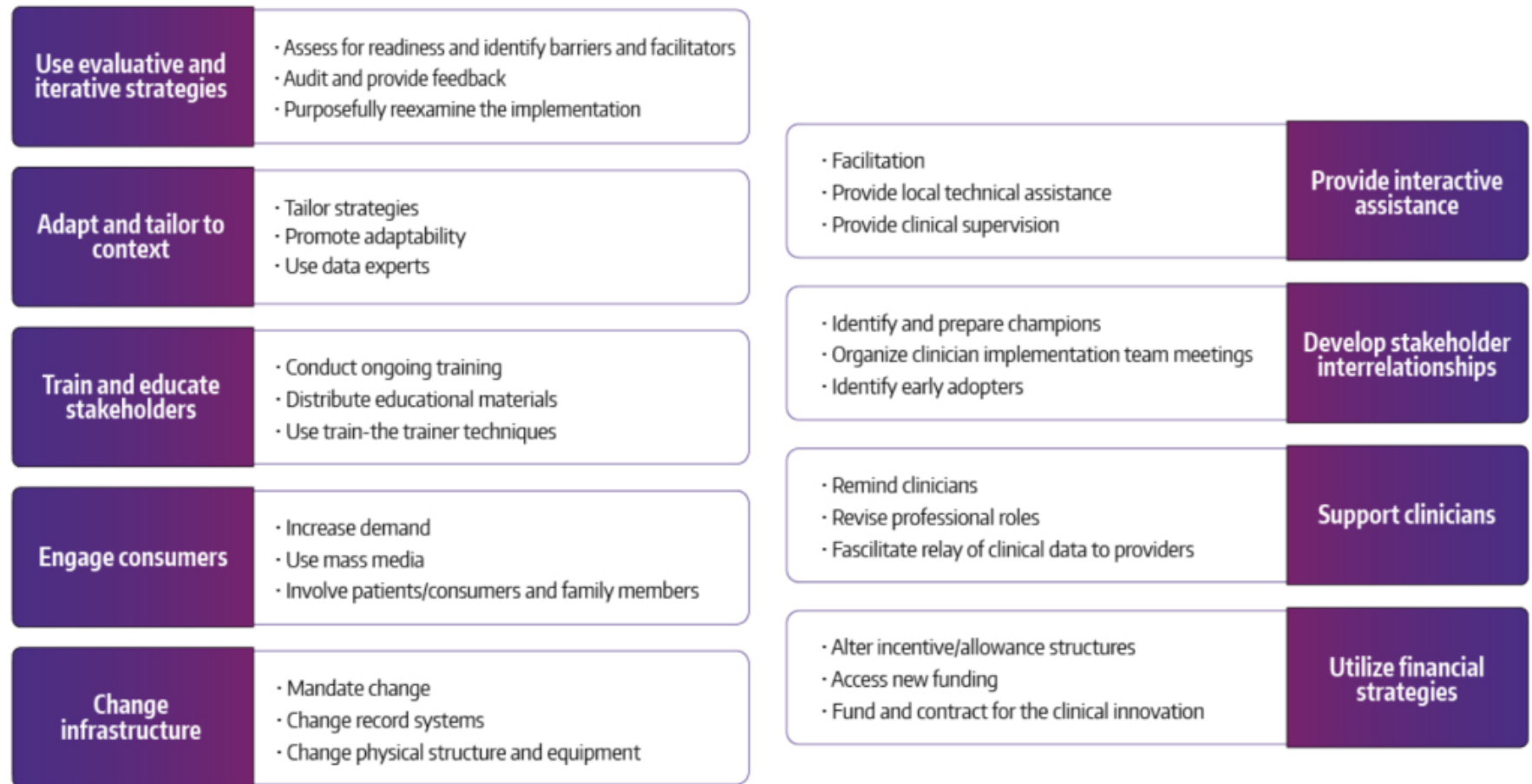
# Step 3: Mix & Bake

Develop implementation strategies



# Step 3: Mix & Bake

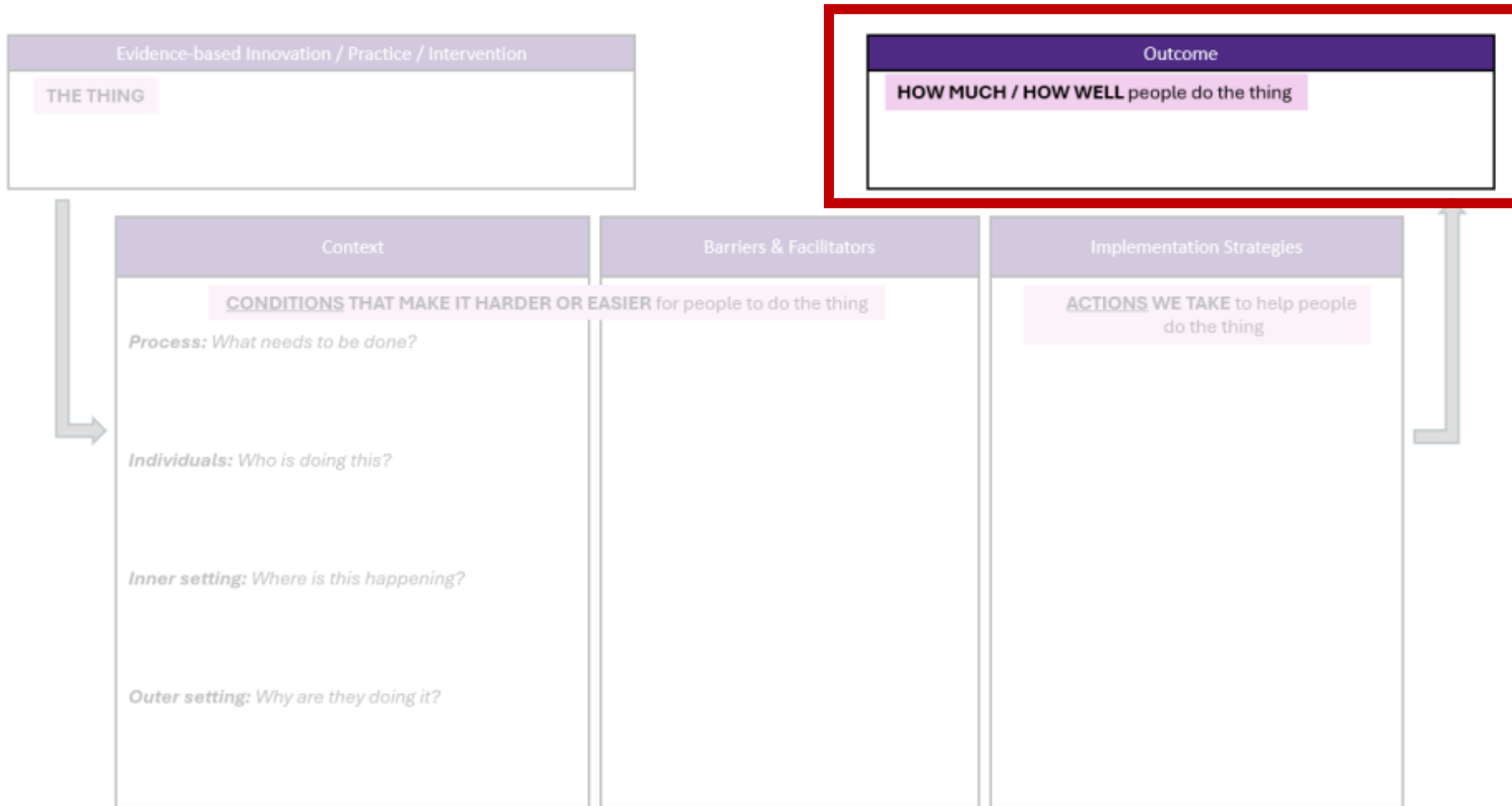
## Create strategies



Adapted from: Waltz, TJ., Powell, BJ., Matthieu, MM. et al. Use of concept mapping to .... *Implementation Sci* 10, 109 (2015) [doi:10.1186/s13012-015-0295-0](https://doi.org/10.1186/s13012-015-0295-0)

# Step 4: Frost and Serve

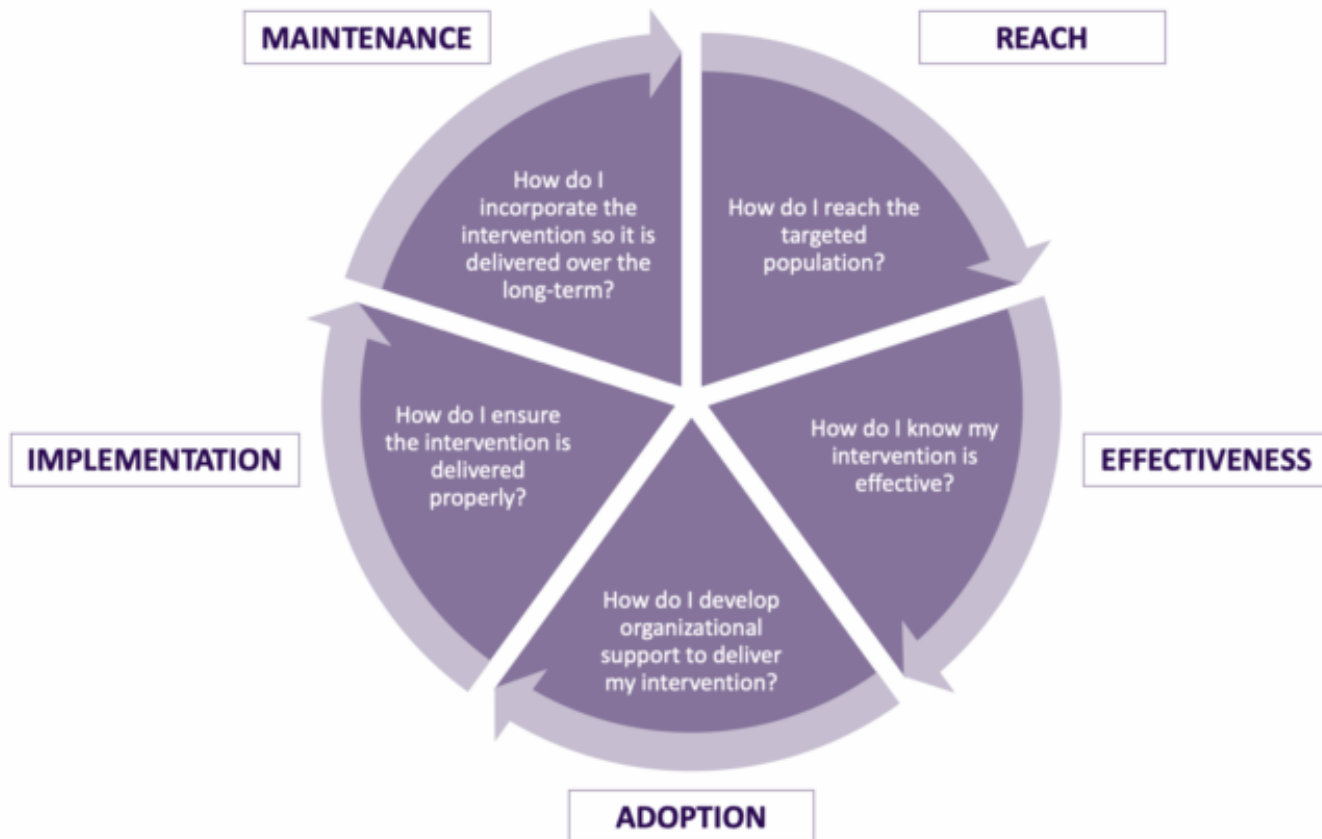
Evaluate **impact**



# Step 4: Frost and Serve

Evaluate **impact**

The **RE-AIM** Framework can be used for evaluating interventions



# Your Implementation Science **RECIPE**

## **Step 1: Decide what to make**

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## **Step 3: Mix & Bake**

- Develop implementation **Strategies**

## **Step 4: Frost & Serve**

- Evaluate **Impact**

# Key Takeaways

**Implementation Science** is a method of understanding how best to help people **DO THE THING**

IS **tools** like the **CFIR** and the **IRLM** provide frameworks for working through implementation problems

These tools can create a **RECIPE** to define the **context**, identify **barriers & facilitators**, create **strategies**, and evaluate **impact** to maximize success

# Resources

**Northwestern Center for Dissemination and Implementation Science –**

**<https://www.feinberg.northwestern.edu/sites/cdis/index.html>**

**Implementation Research Logic Model -**

**<https://cepim.northwestern.edu/implementationresearchlogicmodel>**

**University of Washington Implementation Science Resource Hub -**

**<https://impsciuw.org/>**

**Consolidated Framework for Implementation Science – <https://cfirguide.org/>**

**The Center for Implementation - <https://thecenterforimplementation.com/toolbox/cfir>**

# Thank you!

Please reach out with questions or comments:

[Rebecca.Williams1@nm.org](mailto:Rebecca.Williams1@nm.org)

# Similarities and Differences Between IS and Quality Improvement:

## QI/Improvement Science

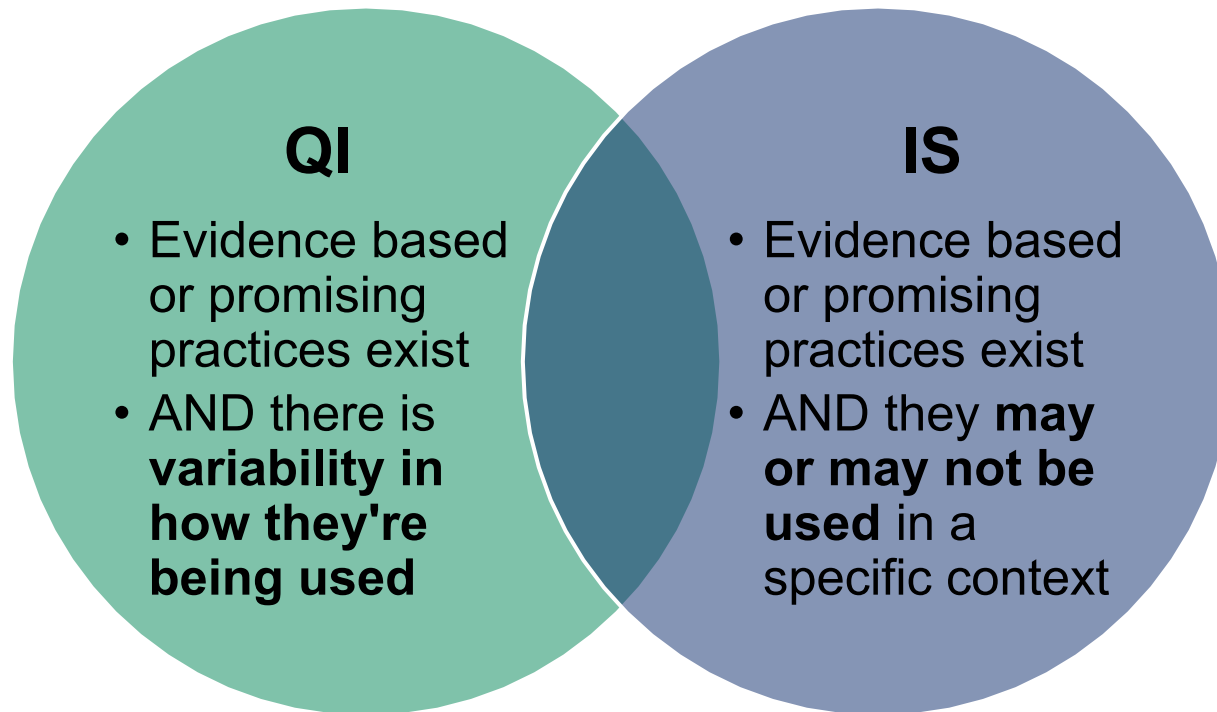


## Implementation Science



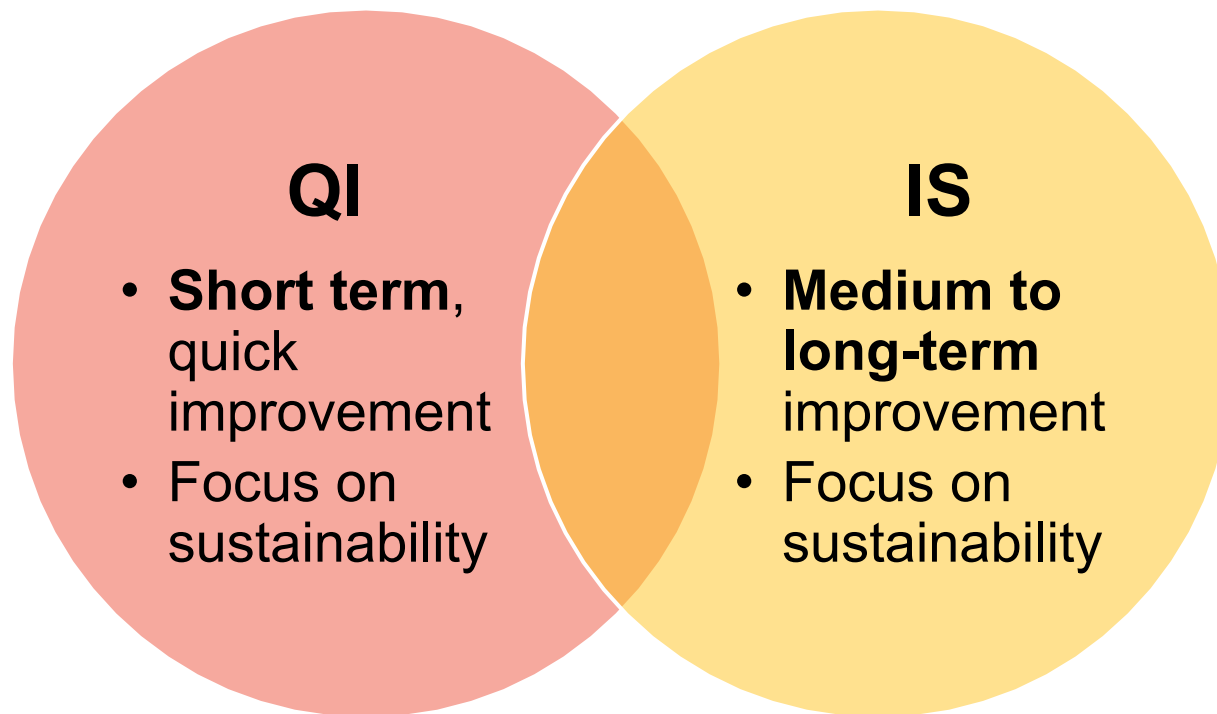
(Nilsen et al., 2022)

# QI versus IS: *Assumptions*



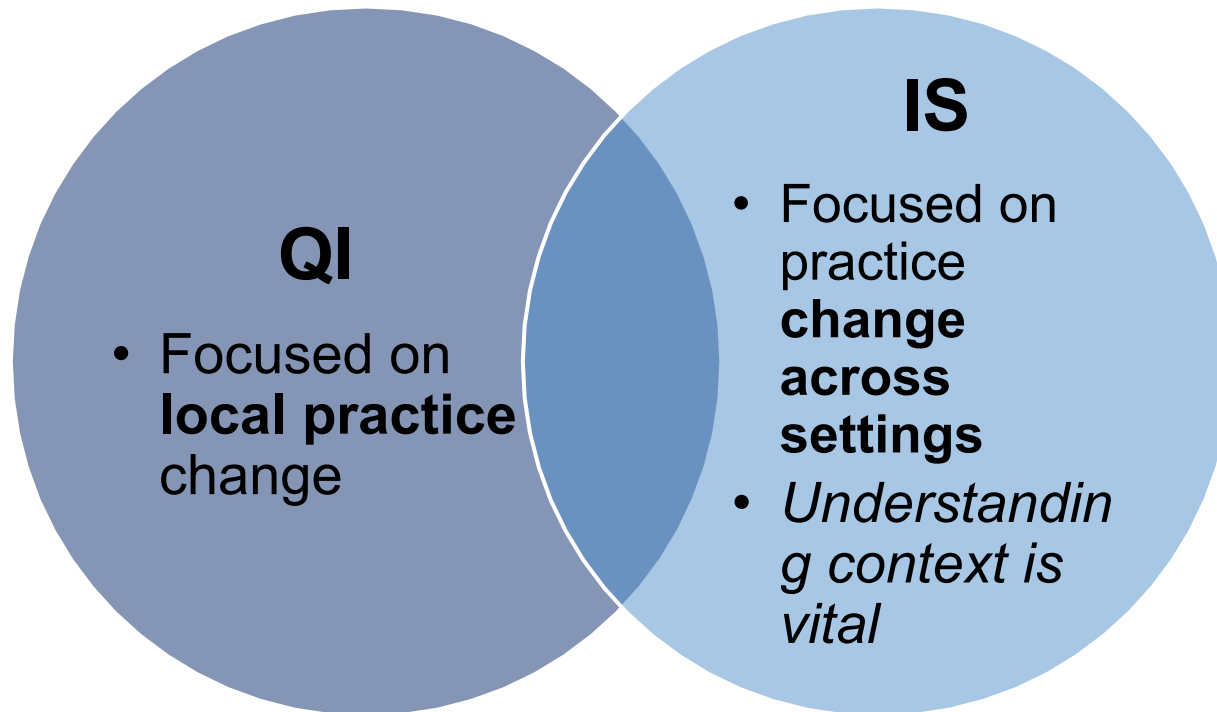
Lane-Fall & Fleisher (2018)

## QI versus IS: *Timeframe*



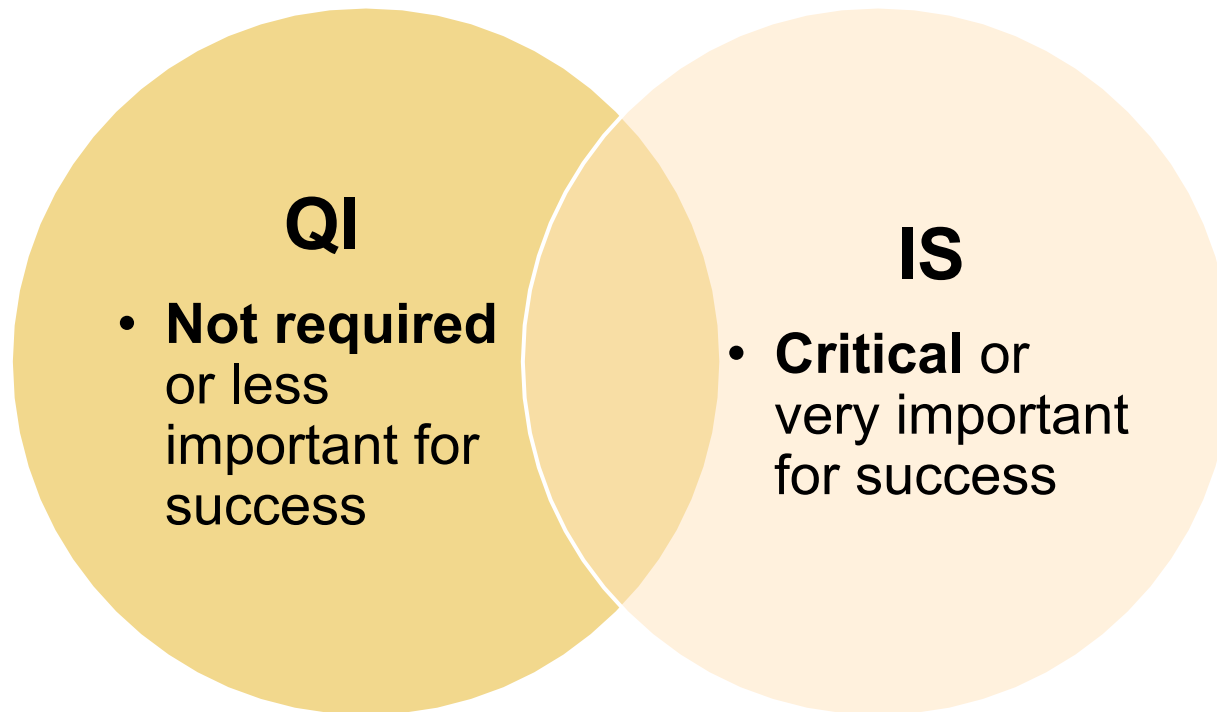
Lane-Fall & Fleisher (2018)

# QI versus IS: *Applicability to Other Settings*



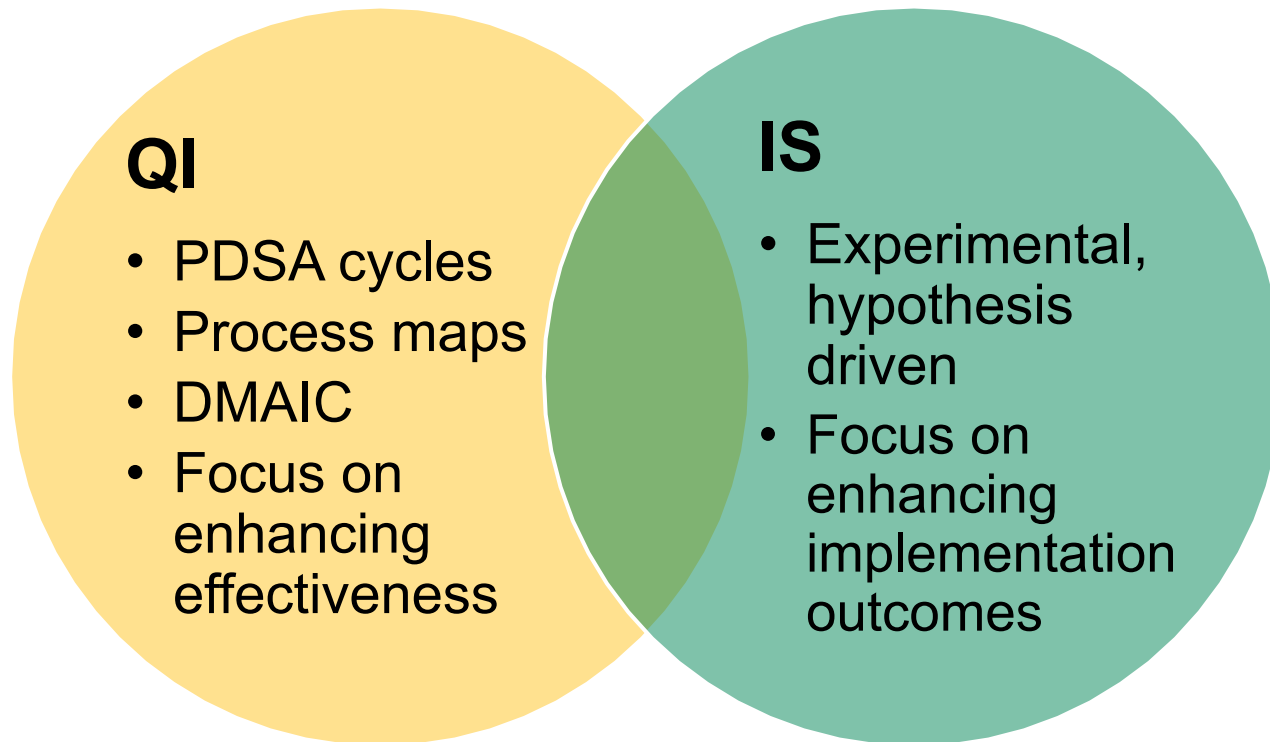
Lane-Fall & Fleisher (2018)

# QI versus IS: *Use of Theories, Models, Frameworks*



Lane-Fall & Fleisher (2018)

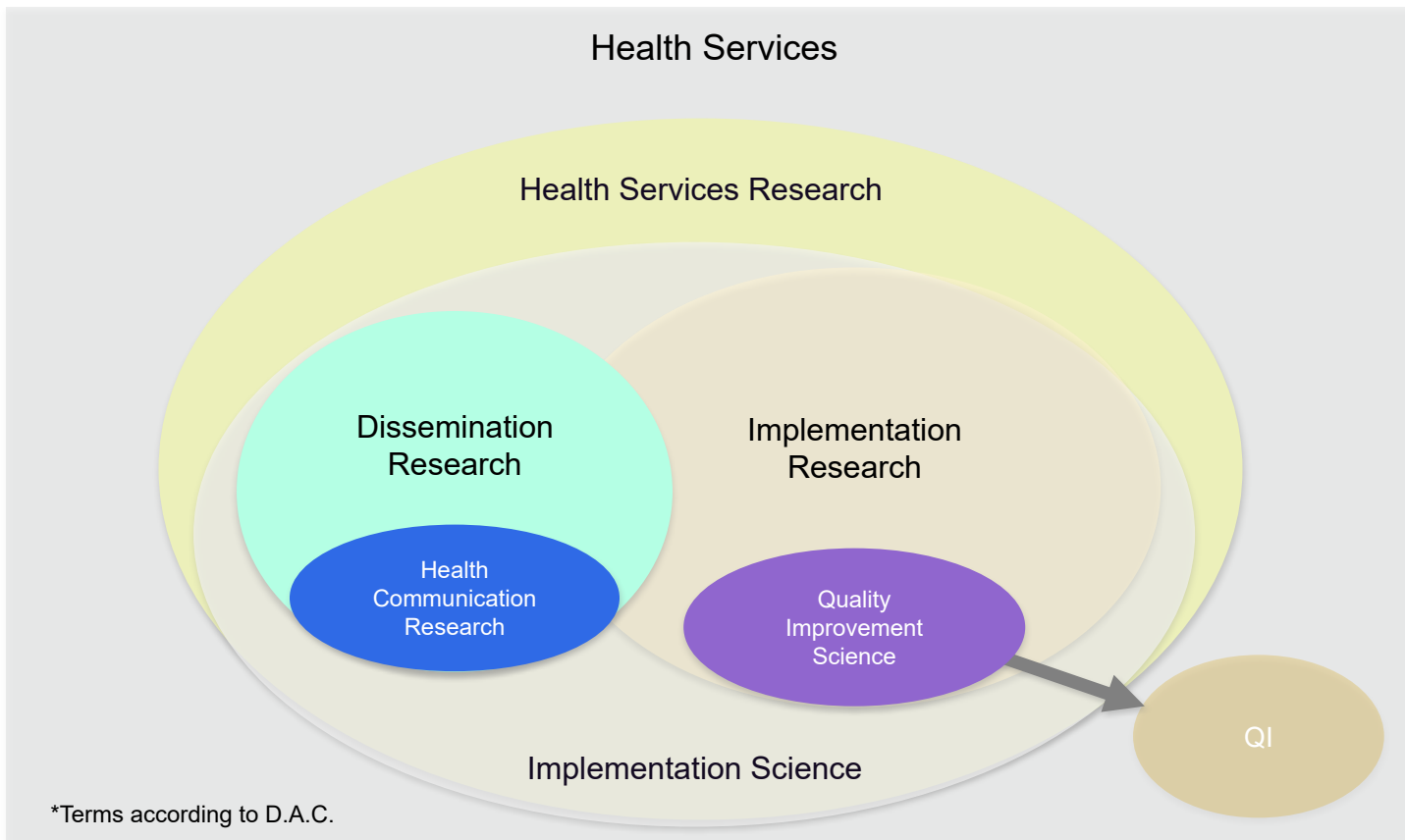
# QI versus IS: *Tools and Outcomes*



Lane-Fall & Fleisher (2018)

# Similarities and Differences Between IS and Quality Improvement:

## *A Big Tent of Terms (and Circles!)*



Institute for Public Health and Medicine  
Northwestern University

# Center for Dissemination & Implementation Science



The Center for Dissemination & Implementation Science (CDIS) is an interdisciplinary center within the Institute for Public Health & Medicine (IPHAM) at Northwestern University Feinberg School of Medicine. CDIS aims to connect public health research and practice by integrating strategies for both patient-focused dissemination and provider- and organization-focused implementation that can be used across diverse service settings and conditions.



Institute for Public Health and Medicine  
Northwestern University

# CDIS/NUCATS Consult Service



Together with our Northwestern collaborators listed below, the Center for Dissemination & Implementation Science coordinates an array of education and research support services. This is an area of active focus for the center, and we will continue to announce new services. We encourage you to [affiliate with the center](#) in order to subscribe to our listserv and stay abreast of our latest offerings.



## D&I Consultations

Dissemination and implementation (D&I) consultation is available to support grants, training and mentorship, and academic research products.

[REQUEST D&I CONSULTATION](#)

Institute for Public Health and Medicine  
Northwestern University

47

# CDIS Annual Reading Course



CDIS and NUCATS jointly offer an annual Implementation Science Reading Course. This course began with 17 students in 2018 and has over 110 students in 2025!

- **96% of attendees would recommend the course to a friend!**
- **90+% of attendees identify as having no or beginner-level experience**
- **87% of attendees want an advanced course in the future**

Center for Dissemination & Implementation Science

# Become a Member to Stay Connected!

CDIS distributes a monthly newsletter that rounds up imp sci training, education, and job opportunities for the NU community.

Prior versions can be viewed on our website.  
Become a member to subscribe!

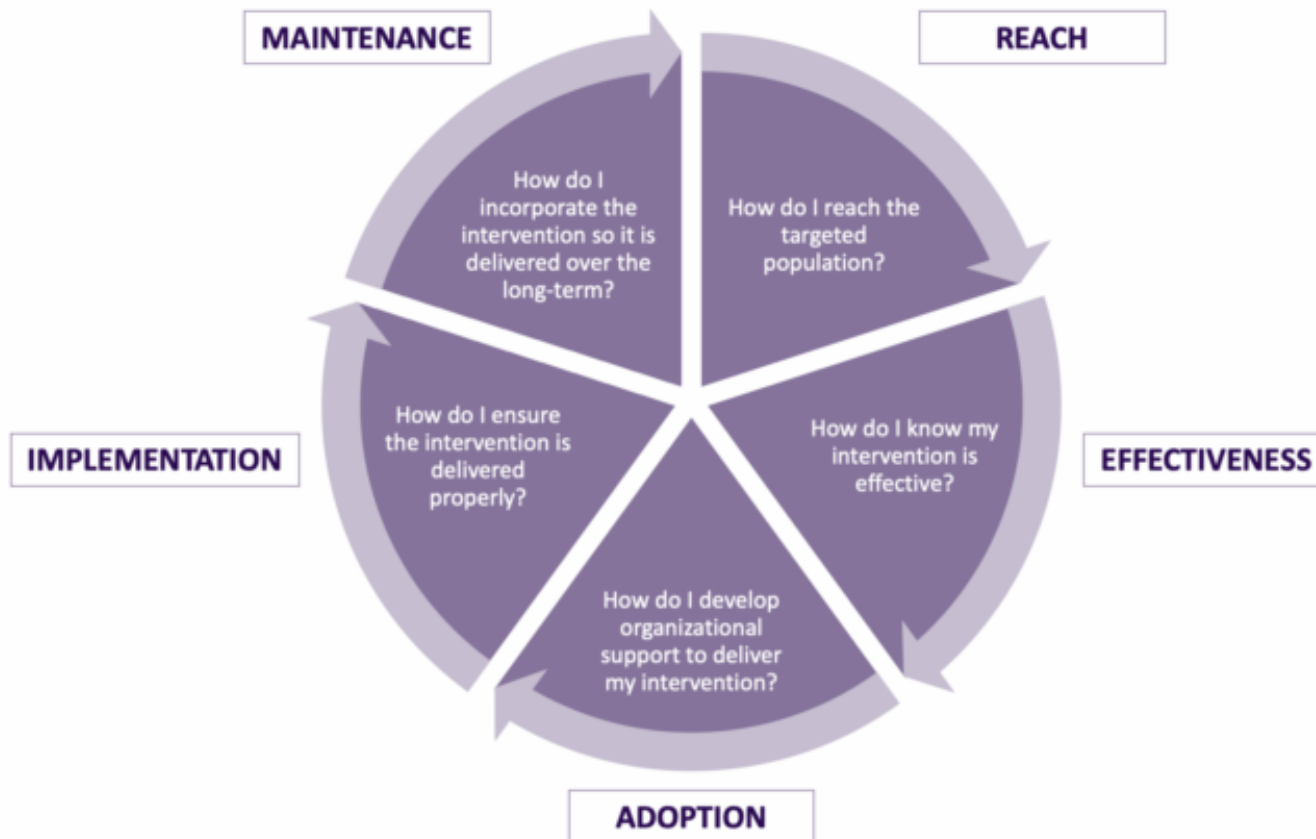


# Extra Slides

# Step 4: Frost and Serve

Evaluate **impact**

The **RE-AIM** Framework can be used for evaluating interventions



## Once more, with less jargon

Evidence-based  
intervention/practice/innovation



**THE THING**

Implementation



**DOING** the thing

Effectiveness research



Looks at whether the thing **WORKS**

Implementation research



Looks at **HOW TO HELP OTHERS** do the thing

Context (determinants)



**CONDITIONS THAT MAKE IT HARDER OR EASIER**  
for others to do the thing (barriers/facilitators)

Implementation strategies



**ACTIONS WE TAKE** to help others do the thing

Implementation / service outcomes



**HOW MUCH / HOW WELL** others do the thing

**Source:** Adapted from Curran, 2020, *Implementation Sci Comm*

# Step 3: Mix & Bake

## Create strategies

- Alter incentive/allowance structures
- Audit and provide feedback
- Build a coalition
- Capture and share knowledge locally
- Centralize technical assistance
- Change physical structure and equipment
- Conduct educational meetings
- Conduct educational outreach
- Conduct ongoing training
- Create a learning collaborative
- Visit other sites
- Develop disincentives
- Develop educational materials
- Develop resource sharing agreements
- Facilitate relay of data to providers
- Identify and prepare stakeholders
- Identify early adopters
- Increase demand
- Mandate change
- Model and simulate change
- Promote adaptability
- Provide ongoing consultation
- Purposefully re-examine implementation
- Recruit, designate and train for leadership
- Remind clinicians
- Shadow other experts
- Stage implementation scale up
- Tailor strategies
- Use advisory boards and workgroups

# The implementation equation



**THE THING** X **IRLM** X **CFIR** X **Work** = **Result**



Based on DiCenso (2009). Image adapted by The Center for Implementation, © 2023 | V2024.01 |  
For full citation: <https://thecenterforimplementation.com/toolbox/implementation-equation>